SymPy Bug Card

Bug 28: solveset returns -1 for a principal cube-root equation

Task. Solve the principal cube-root equation over the real numbers.

$$x^{1/3} = -1$$

SymPy's answer.

$$\mathrm{solveset}(\mathrm{Eq}(x^{1/3}, -1), x, \mathbb{R}) = \{-1\}.$$

Correct answer.

$$\varnothing$$

Explanation. Substituting SymPy's returned value gives $(-1)^{1/3} + 1 \neq 0$. SymPy's $x^{1/3}$ is a principal complex power, so $(-1)^{1/3} = \exp(i\pi/3) = 1/2 + \sqrt{3}i/2$, not the real cube root -1.

Likely root cause. In `sympy/solvers/solveset.py`, `_invert_real` handles odd-denominator rational powers using `real_root` semantics, and the later domain check verifies definedness rather than equation truth. The deduplication assessment frames this as a new instance of a known failure mode.

Artifact (GitHub).
`bug-report/bug-028-solveset-returns-1-for-a-principal-cube-root-equation/`